

Data Analyst Training Programme

Stored Procedures & Views

Introduction to Database Objects

Database Abstraction Layers

- Views provide virtual tables for simplified data access
- Stored procedures encapsulate reusable business logic
- Enhanced security through controlled data exposure
- Improved performance through pre-compiled execution

Enterprise Benefits

- Code standardisation across applications
- Centralised business rule enforcement
- Reduced network traffic and improved performance
- Simplified maintenance and version control

Development Workflow

- Views for complex query abstraction
- Procedures for repetitive analytical tasks
- Functions for reusable calculations
- Triggers for automated data processing

Views - Database Abstraction and Security

View Types and Application

- Simple views for single table abstractions
- Complex views joining multiple tables
- Materialised views for performance optimisation
- Updatable views for controlled data modification

Security and Access Control

- **Column-level security** through selective exposure
- Row-level security with **WHERE** conditions
- User-specific views for **role-based access**
- Sensitive **data masking** and **transformation**

Performance Considerations

- Query optimisation through view design
- Index utilisation in underlying tables
- Avoiding nested view dependencies
- Materialised views for heavy computations

Stored Procedures - Encapsulating Business Logic

Stored Procedure Advantages

- Pre-compiled execution for improved performance
- Parameter-driven flexible business logic
- Transaction control and error handling
- Reduced network traffic through server-side processing

Parameter Types

- **IN** parameters for input values
- **OUT** parameters for return values
- **INOUT** parameters for bidirectional data
- Default parameter values for flexibility

Business Logic Applications

- Complex data validation routines
- Multi-step analytical calculations
- Automated report generation
- Data transformation pipelines

Advanced Stored Procedure Techniques

Control Flow Structures

- Conditional logic with **IF-THEN-ELSE** statements
- Iterative processing with WHILE loops
- Cursor-based row-by-row processing
- Dynamic SQL generation and execution

Error Handling and Transactions

- Exception handling with DECLARE handlers
- Transaction management for data consistency
- Rollback mechanisms for error recovery
- Custom error messages and logging

Performance Optimisation

- Efficient cursor usage patterns
- Temporary table strategies
- Bulk processing techniques
- Query plan optimisation

Functions and Triggers for Automation

User-Defined Functions

- Scalar functions returning single values
- Table-valued functions returning result sets
- Deterministic vs non-deterministic functions
- Function security and performance considerations

Trigger Applications

- Automatic audit trail creation
- Data validation and business rule enforcement
- Derived field calculations and updates
- Integration with external systems

Automation Strategies

- Event-driven data processing
- Automatic report scheduling
- Data quality monitoring
- Performance metric tracking

Assignment

Complete these stored procedure and view challenges using the Northwind database:

View Creation and Management

- Create a sales dashboard view combining order data with customer and product information
- Build a security-focused view that provides regional managers access only to their geographic data
- Design a performance monitoring view that tracks key business metrics over time

Stored Procedure Development

- Develop a flexible reporting procedure that accepts parameters for date ranges, countries, and product categories
- Create a customer segmentation procedure that automatically categorises customers based on purchasing patterns
- Build a data quality validation procedure that identifies and reports inconsistencies across related tables

Functions and Automation

- Write a function to calculate profit margins based on product costs and selling prices
- Create a function that determines the optimal reorder quantity for products based on historical sales
- Develop triggers to maintain audit trails for critical data changes

Advanced Integration Projects

- Design a complete analytical framework combining views, procedures, and functions for monthly business reporting
- Create a customer lifecycle management system using stored procedures for automated tier assignments and communications
- Build a performance monitoring system that uses triggers to automatically track and alert on key business metrics

End of the training...

See you around, **[student name]**.

Stored procedures and views represent the foundation of enterprise database architecture, enabling code reusability, enhanced security, and improved performance through centralised business logic. You now possess the essential skills to create maintainable, scalable database solutions that support sophisticated analytical requirements whilst ensuring data integrity and access control across any enterprise environment.

Thank you, [student name].

Any Questions?